

CS 137, Deep Neural Networks

HW 2, SkyOracle

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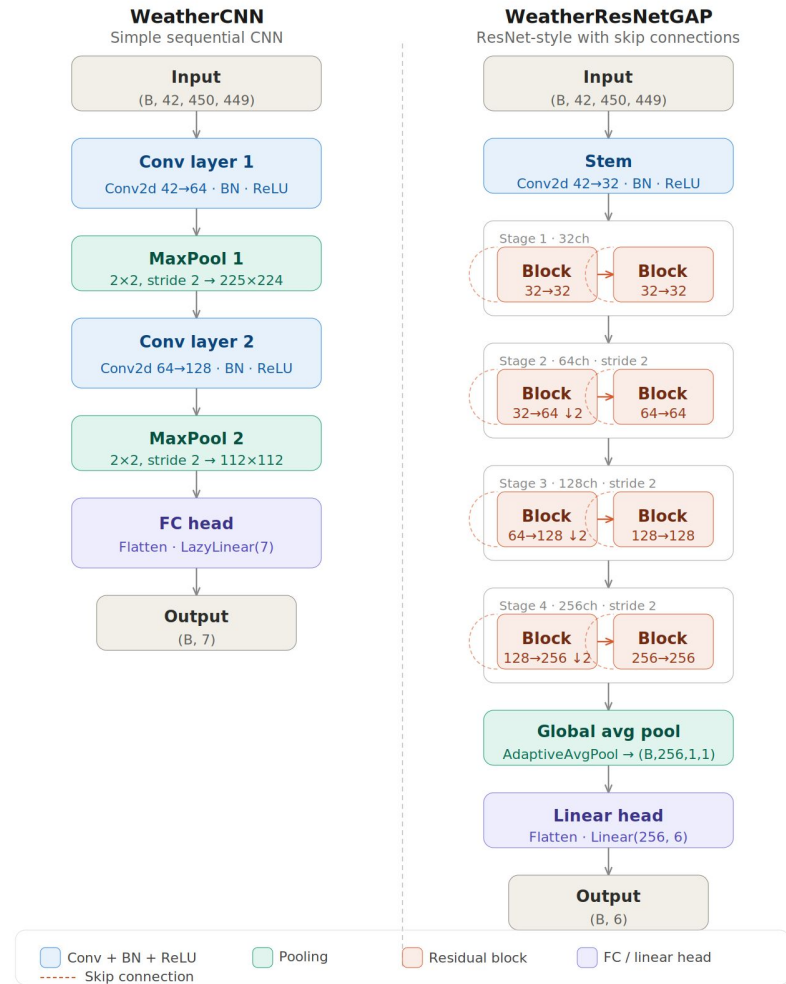


Model Architectures

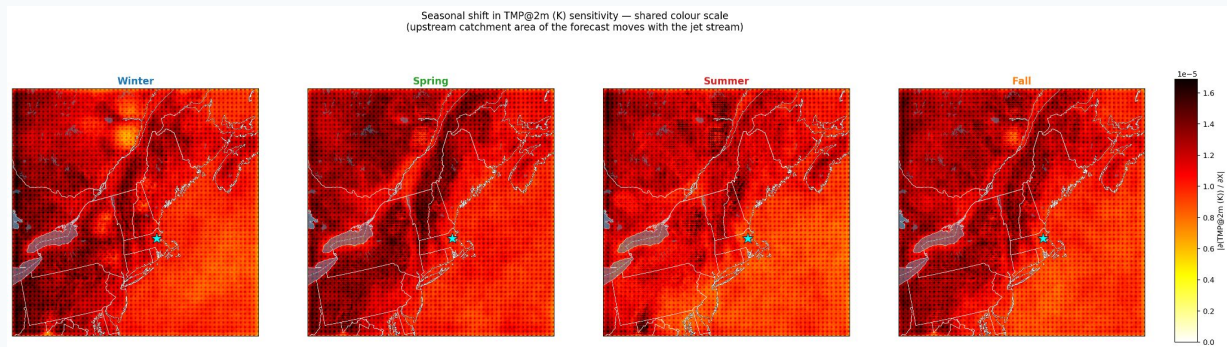
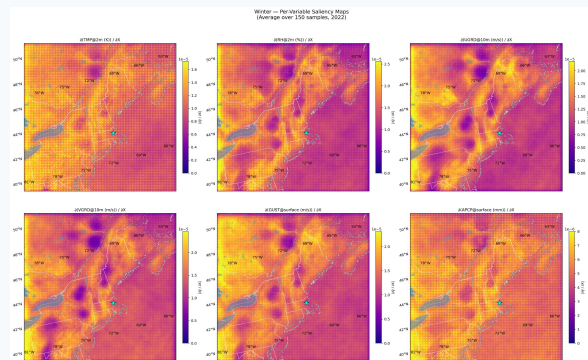
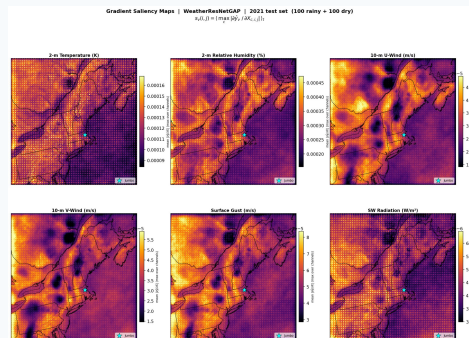
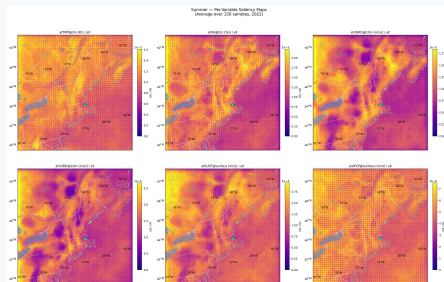
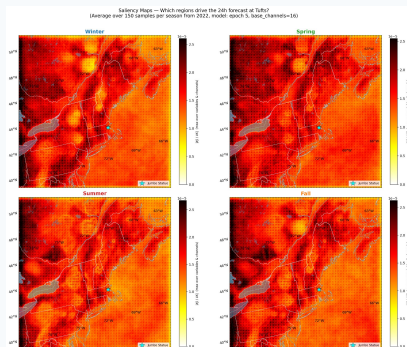
WeatherCNN (left) is a classic flat CNN, two conv blocks each followed by max-pooling to halve spatial dimensions, then a single fully-connected head outputting 7 values.

WeatherResNetGAP (right) is considerably deeper, a stem layer feeds into 4 progressive stages (32→64→128→256 channels), each containing two **BasicBlock** residual units with skip connections (shown as dashed arcs). Stride-2 downsampling replaces max-pooling, and a global average pool collapses spatial dimensions before the linear head, outputting 6 values.

Bonus, WeatherResNetGAP also trained that compares the above version with the seventh feature included. However the batch job for training the model is frozen on the HPC side :(



Diagnostics





Independent study

Feature ablation study done by two methods:

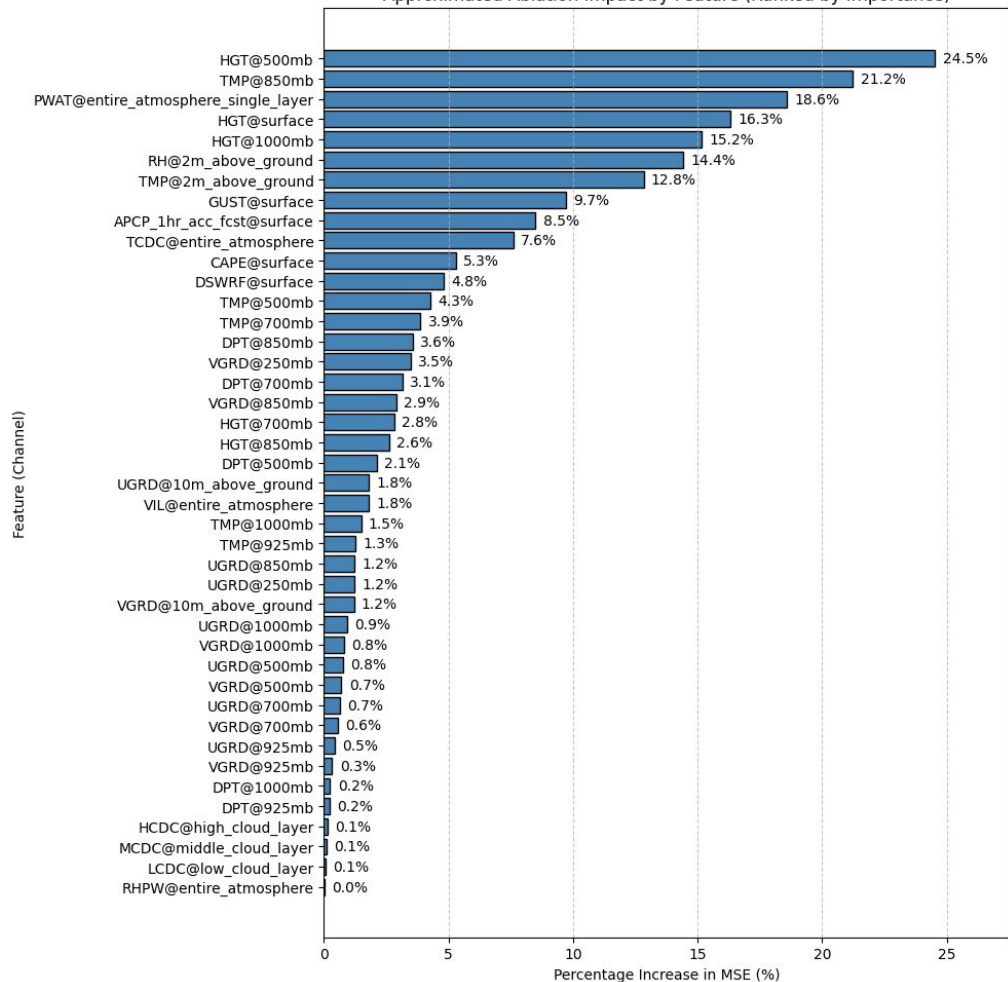
- Turn off one input channel at a time (of 42) and measure how much MSE goes up vs the full model.

What matters most: Height (HGT) and temperature (TMP) dominate—especially 500 mb HGT and 850 mb TMP; many wind and cloud channels barely move the error.

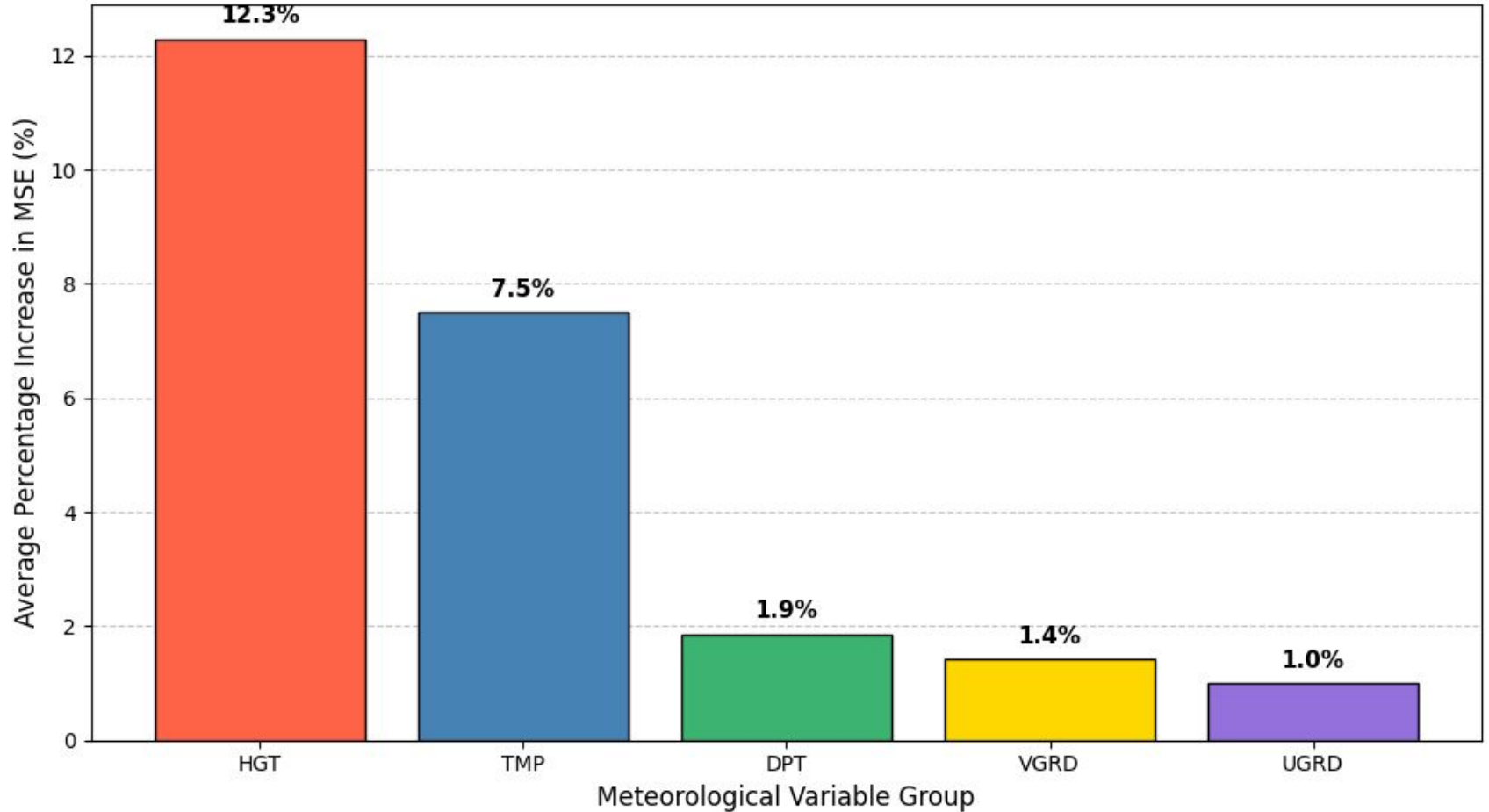
Vertical story: Sensitivity varies by pressure level (e.g. HGT key aloft + near surface, moisture mid-low troposphere), the model leans on mass & thermal structure, not every field equally.

- two different versions of WeatherResNetGAP, one with and without the seventh feature to see the differences in error loss metrics

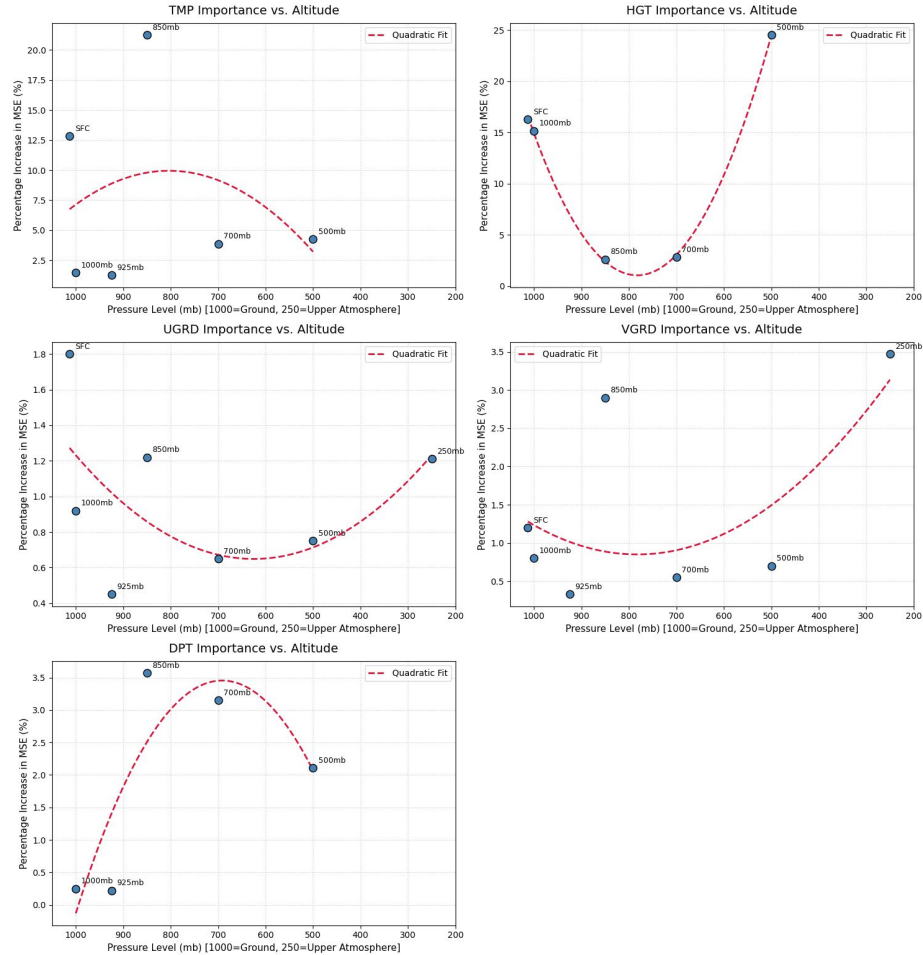
Approximated Ablation Impact by Feature (Ranked by Importance)



Average Feature Importance by Variable Group



Variable Importance Tracking Through the Atmospheric Profile





Questions?